

PREDICTING WHEN LoRA MERGING FAILS: A RATE-DISTORTION VIEW

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ABSTRACT

Merging LoRA adapters into one deployable model is judged empirically: methods are compared on benchmarks with no notion of how much room is left to improve, and a method that wins on one base model can become the worst on another with no warning. We address both problems by treating merging as lossy compression: how well can T adapters be represented by a single model, judged by the task that suffers most? This rate-distortion view yields a closed-form lower bound on the achievable worst-task loss (an irreducible floor plus a compression cost) and an explicit encoder that matches it up to a logarithmic factor in T . Two practical tools follow. On real adapters the exact encoder fails for a measurable spectral reason, but a one-line ridge correction repairs it; the repaired method, RD-ENCODER RIDGE, attains the lowest worst-task loss of the ten methods we benchmark, on all four base models, at default hyperparameters, with a lead that widens as more tasks are merged. The same analysis gives a CPU-only, minutes-long audit of trained adapters that predicts when the popular TIES method flips from best to worst; we pre-registered its prediction for a third base (Mistral-7B) before the experiment ran, and the prediction held. On every adapter collection we measure, the floor is zero: the loss merging adds is avoidable in principle, and our encoder recovers a substantial fraction of it without retraining ($0.22 \rightarrow 0.094$ on Llama-3.1, 3-seed mean). Loss rankings also predict downstream accuracy rankings in seven of eight model-metric pairs.

1 INTRODUCTION

Model merging is everywhere, but nobody knows the limit. Low-Rank Adaptation (Hu et al., 2022) has made it routine to maintain dozens of task-specific fine-tunes per base model; *merging* them into a single deployable model is now the dominant practical question, with Task Arithmetic (Ilharco et al., 2023), TIES (Yadav et al., 2023), DARE (Yu et al., 2024), KnOTS (Stoica et al., 2025), Task-Vector Quantization (Kim et al., 2025), and many successors competing on it. Two operational questions go unanswered: *how much room is left* (a benchmark score says nothing about whether a better merger could still halve its error) and *when a method will fail* (a merger that wins on one base can end up the worst on another with no warning). Both reduce to a question no method addresses: at a given storage budget, what is the best *possible* merging algorithm, and how far from it are today’s methods?

Contribution in one sentence. *From a worst-task rate-distortion analysis of LoRA merging we obtain two tools: a merging method (RD-ENCODER RIDGE) with the lowest worst-task loss of the ten methods we benchmark at default hyperparameters on every base we test, and a CPU-only audit of any set of trained adapters, taking minutes and no GPU, that predicts when classical methods such as TIES flip from best to worst; its predictions held on all three bases we tested, one of them pre-registered¹.*

The lens: merging as compression. Both tools come from treating merging as lossy compression: T task-specific weight updates must be encoded into one model within a storage budget, and

¹Throughout, *pre-registered* means the prediction was committed to our timestamped, version-controlled experiment log *before* the corresponding compute was run; an anonymized bundle reproducing the commit trail is included in the supplementary material.

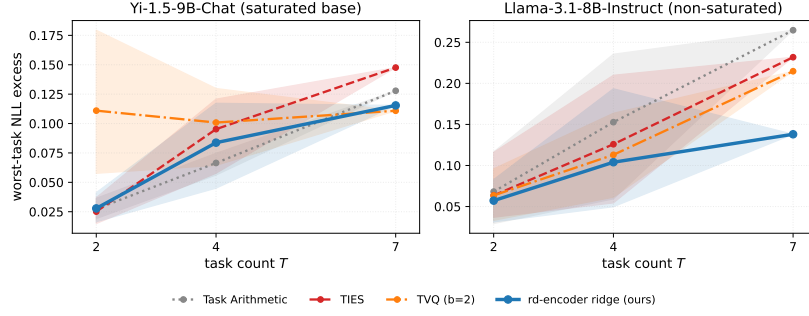


Figure 1: **Same merge methods, opposite outcomes across bases.** Worst-task NLL excess vs. task count $T \in \{2, 4, 7\}$ on three bases, ordered by R , the spread in per-task sign agreement that our audit measures: saturated Yi-1.5-9B-Chat ($R = 0.077$), pre-registered Mistral-7B-v0.3 ($R = 0.0326$), non-saturated Llama-3.1-8B-Instruct ($R = 0.028$); bands are per-subset min/max. Yi, above the predicted window $[0.025, 0.075]$, inverts: TIES becomes *worst* at $T = 7$ (§6.4; threshold in Appendix I); Mistral, inside the window, stays neither best nor worst, exactly as pre-registered. RD-ENCODER RIDGE (ours) improves as tasks are added: its excess falls from $0.84\times$ to $0.52\times$ Task Arithmetic’s over $T = 2 \rightarrow 7$ on Llama, and from $1.38\times$ to $0.47\times$ on Mistral.

quality is the extra held-out loss (negative log-likelihood, NLL) paid by the *worst*-affected task. For this problem we prove a rate-distortion theorem: lower and upper bounds that match in their rate exponent and split the worst-task error into an **irreducible floor**, set entirely by how much the task subspaces overlap and beyond the reach of any encoder at any bit budget, plus a **compression term** that decays exponentially in the bit budget and that an explicit encoder achieves up to a constant. Measuring the overlap on 16 trained LoRA adapters (four bases $\times$ four tasks) settles which regime real merging occupies: task subspaces are linearly *independent* in every layer of every model, so the floor is exactly zero and the 0.10–0.22 nats/token the five standard methods leave on the table is recoverable algorithmic slack, not an information-theoretic cost. Zero overlap also predicts which methods can work: subspace-alignment merging (KnOTS (Stoica et al., 2025)) has nothing to exploit and is statistically indistinguishable from naive averaging, while TIES, which attacks sign-level interference instead, is the only baseline that separates. What §6 validates on real LLMs is thus the regime *diagnosis*, not the quantitative rate-distortion curve.

Contributions.

1. **A merge method and an adapter audit, both falling out of the bound (§6).** RD-ENCODER RIDGE, the encoder from our achievability proof repaired by a single ridge term after a measured spectral failure, has the lowest worst-task excess of all ten methods at default hyperparameters on every base we test (a tuned RegMean edges it on one saturated base; §6.3). A perturbation analysis of the TIES sign election yields a CPU-only audit and a five-rule decision tree that predict when classical mergers flip from best to worst, with the window’s upper edge and its ambiguous interior confirmed on three bases (one pre-registered).
2. **A rate-distortion theorem for worst-task merging (§4–§5).** A reduction identity (Lemma 1) splits worst-task distortion into a quantization term around the H -weighted centroid plus a spread term no encoder can remove; under the random-subspace hard distribution the spread term equals $B^2(1 - d_{\text{eff}}/(Tr))$ exactly (Lemma 2). At rate R , no encoder/decoder pair beats this floor plus $c_{\text{TQ}}(B^2 d_{\text{eff}}/(Tr)) 2^{-2R/d_{\text{eff}}}$, for arbitrary positive-definite per-task spectra (Theorem 3). In the floor-zero regime an explicit encoder (randomized orthogonal mixing followed by scalar quantization) achieves the same $2^{-2R/(Tr)}$ decay, so $\mathcal{D}^* = \Theta(B^2 2^{-2R/(Tr)})$ for fixed T (Theorem 4); the measured *logarithmic* growth of the constant in T is posed as a sharp open problem (Remark 5).
3. **A real-LLM calibration that turns the bound into a measured gap (§6).** Floor zero everywhere; 0.10–0.22 nats/token of recoverable headroom; and a universal “less-is-more” dip at 2-bit task-vector quantization on all four architectures.

Scope. The bound certifies a *lower envelope*, not a tight per-instance grade: real task vectors are more benign than the worst case, so a method’s measured gap to $\mathcal{D}^*$ is an upper estimate of its true suboptimality. The matching upper bound is proved in the generic random-subspace regime with $Tr \leq d$; the floor-positive regime has an exponent gap we leave open (Remark 6). §3 formalizes the model; §4 proves the lower bound; §5 gives the matching construction; §6 reports the empirical studies; §7 discusses the cross-entropy bridge and limitations.

2 RELATED WORK

Model merging: constructions without limits. A wave of post-hoc merging methods followed Task Arithmetic (Ilharco et al., 2023): Fisher-weighted averaging (Matena & Raffel, 2022), sparsification approaches (TIES (Yadav et al., 2023), DARE (Yu et al., 2024), DELLA (Deep et al., 2024)), rotation- and subspace-alignment families (KnOTS (Stoica et al., 2025), TSPA (Li et al., 2025), DO-Merging (Zheng et al., 2025), Core Space (Panariello et al., 2025), TARA (Jeong et al., 2026), ARM (Yao et al., 2026)), and merge-time quantization (TVQ (Kim et al., 2025), 1-bit-Merging (Liu et al., 2025a)). All are *constructions*: each proposes an encoder and evaluates it empirically; none establishes a fundamental limit on merging quality at a given storage budget. This paper supplies that limit and, in the generic regime, a matching construction.

Existing theoretical analyses of merging. *Weight disentanglement* (Ortiz-Jimenez et al., 2023) identifies the rank- r Hessian structure of LoRA fine-tunes and provides the $H_t = U_t D_t U_t^\top$ formalism we adopt, but gives no rate-distortion bounds. Task-Vector Bases (Zeng et al., 2025) posits a thin-Gaussian-shell geometry; ATM (Zhou et al., 2024) and its successor (Zhou et al., 2025) develop a second-order Taylor bound on task-arithmetic error. None frames merging as lossy source coding; to our knowledge we are the first to pose merging as a multi-source rate-distortion problem with worst-task distortion as the figure of merit.

Closed-form merging lineage. Our encoder sits in a line of closed-form merges that invert a per-task curvature surrogate. RegMean (Jin et al., 2023) solves a least-squares system in the per-task input-activation Grams (data-dependent, forward passes required); LoRM (Wen et al., 2024) extends it to low-rank adapters and RegMean++ (Park et al., 2025) refines the regularization. Ours differs in three ways: it is *data-free* (the projector comes from the adapter factors alone); it is derived from a worst-task rate-distortion lower bound rather than an empirical fit; and its Tikhonov λ is the numerical-conditioning floor of Theorem 4 exposed as a single hyperparameter. We benchmark against a data-free RegMean in §6.3 and sweep its ridge λ in Appendix G so the comparison does not conflate the centroid with the regularizer.

Rate-distortion and rotation-for-quantization ancestry. Classical rate-distortion theory (Shannon, 1959; Cover & Thomas, 2006) covers the single-source setting; multi-description coding (El Gamal & Cover, 1982) treats *average*, not maximum, distortion, which Lemma 1 shows is strictly weaker. The direct ancestor of our achievability construction is TurboQuant (Zandieh et al., 2025), whose Theorem 3 our Theorem 4 recovers at $T = 1$. Structured rotations as quantization pre-conditioners (QuIP (Tseng et al., 2024), QuaRot (Ashkboos et al., 2024), SpinQuant (Liu et al., 2025b), RoLoRA (Huang et al., 2024)) motivate our randomized orthogonal mixers but are single-source.

A recent negative result, and why a lower bound still matters. A large-scale empirical study (Hitit et al., 2025) reports that across four LLM families only Task Arithmetic reliably produces constructive interference, with subspace-alignment methods often failing. Their result is an *upper bound* on specific algorithms; our Theorem 3 is a *lower bound* on every algorithm, and our measurements reconcile the two: real LoRA task subspaces are linearly independent ($d_{\text{eff}} = Tr$ in every layer of all four bases), so subspace alignment has no overlap to exploit (we measure KnOTS indistinguishable from Task Arithmetic) while sign-level TIES helps. Where our ordering differs from theirs, the difference is one of metric and setup (average interference across 16 heterogeneous benchmarks vs. worst-task NLL excess on a controlled matrix), not a contradiction.

3 PROBLEM SETUP AND PRELIMINARIES

Task vectors and per-task geometry. Fix integers $T \geq 1, d \geq 2, 1 \leq r \leq d$, and a radius $B > 0$. Each of the T tasks is specified by a pair $(\tau_t, H_t) \in \mathbb{R}^d \times \mathbb{R}^{d \times d}$ with

$$H_t \succeq 0, \quad \text{rank}(H_t) \leq r, \quad \tau_t \in V_t := \text{range}(H_t), \quad \tau_t^\top H_t \tau_t \leq B^2. \quad (1)$$

Per-task loss is the quadratic $f_t(w) := \frac{1}{2}(w - \tau_t)^\top H_t(w - \tau_t)$. We write $\boldsymbol{\tau} := (\tau_t)_{t=1}^T$, $\mathbf{H} := (H_t)_{t=1}^T$, and call tuples satisfying (1) *admissible*. The specialization $H_t = P_{V_t}$ recovers the weight-disentangled LoRA setting of Ortiz-Jimenez et al. (2023); general $H_t \succeq 0$ subsumes Fisher-metric surrogates for cross-entropy.

Worst-task distortion. A merged model $w^* \in \mathbb{R}^d$ incurs per-task distortion $D_t(w^*) := (w^* - \tau_t)^\top H_t(w^* - \tau_t)$; the figure of merit is the *worst-task distortion* $\Delta(w^*; \boldsymbol{\tau}, \mathbf{H}) := \max_{t \in [T]} D_t(w^*)$: a merged model that collapses on one task is not deployable, and classical multi-description coding (El Gamal & Cover, 1982) treats average distortion, which has a strictly weaker rate-distortion function (Lemma 1).

Rate- R merging codes. A rate- R merging code is a pair (E, D) with encoder $E : (\mathbb{R}^d)^T \rightarrow \{1, \dots, 2^R\}$ and decoder $D : \{1, \dots, 2^R\} \rightarrow \mathbb{R}^d$, both of which may depend on $\mathbf{H}$ as shared side information: in deployment $\mathbf{H}$ is determined by the LoRA factors both ends already hold, so only the R bits describing $\boldsymbol{\tau}$ are charged to the rate. Our lower bound holds even when $\mathbf{H}$ is revealed for free to both ends. The *rank- r rate-distortion function* is

$$\mathcal{D}^*(T, d, r, B, R) := \inf_{(E, D)} \sup_{(\boldsymbol{\tau}, \mathbf{H}) \text{ admissible}} \mathbb{E}[\Delta(w^*; \boldsymbol{\tau}, \mathbf{H})], \quad w^* := D(E(\boldsymbol{\tau})), \quad (2)$$

with the expectation over any shared randomness. In the floor-zero Stiefel regime the paper determines $\mathcal{D}^*$ up to a universal constant: a lower bound for every code (Theorem 3) and an explicit code matching it (Theorem 4).

Key derived quantities. We repeatedly use

$$\bar{H} := \frac{1}{T} \sum_{t=1}^T H_t, \quad \bar{\tau}_H := \bar{H}^+ \cdot \frac{1}{T} \sum_{t=1}^T H_t \tau_t, \quad d_{\text{eff}} := \text{rank}\left(\sum_{t=1}^T P_{V_t}\right), \quad (3)$$

where $(\cdot)^+$ is the Moore–Penrose pseudoinverse. The H_t -weighted centroid $\bar{\tau}_H$ is the unique element of $\text{range}(\bar{H}) = V_1 + \dots + V_T$ with $\bar{H} \bar{\tau}_H = T^{-1} \sum_t H_t \tau_t$, reducing to the arithmetic mean when $H_t = I_d$; the *effective dimension* $d_{\text{eff}} \leq \min(Tr, d)$ counts the independent directions the decoder must reconstruct.

Hard distribution P^* . The lower bound is established against a distribution P^* on admissible tuples. For each task t , independently: (i) sample $U_t \in \text{Stiefel}(d, r)$ uniformly and set $V_t := \text{range}(U_t)$; (ii) fix a positive-definite diagonal $D_t \succ 0$ (the task- t spectrum) and set $H_t := U_t D_t U_t^\top$; (iii) sample $u_t \sim \text{Unif}(S^{r-1})$ and set $\tau_t := U_t B D_t^{-1/2} u_t$. The normalization $\tau_t^\top H_t \tau_t = B^2$ holds by construction, and the $D_t^{-1/2}$ stretch makes the second-moment identity $\mathbb{E}[H_t \tau_t \tau_t^\top H_t \mid U_t] = (B^2/r) H_t$ hold *regardless of* D_t , the feature that makes Theorem 3 apply to arbitrary task-dependent spectra.

Yao’s minimax bridge. By Yao’s minimax principle (Yao, 1977), $\mathcal{D}^*$ is at least the Bayesian rate-distortion function under the prior P^* :

$$\mathcal{D}^*(T, d, r, B, R) \geq \inf_{(E, D)} \mathbb{E}_{(\boldsymbol{\tau}, \mathbf{H}) \sim P^*} [\Delta(w^*; \boldsymbol{\tau}, \mathbf{H})] \quad (4)$$

All lower bounds below are proved against the right-hand side.

4 LOWER BOUND

The lower bound rests on three ingredients: a max- $\geq$ -avg identity converting max-distortion into average distortion plus an irreducible floor; a closed-form evaluation of that floor under P^* ; and a

Shannon converse on the remaining compression subproblem, conditional on the curvature data $\mathbf{H}$. Full proofs are in the supplementary material.

Lemma 1 (H_t -weighted max $\geq$ avg identity). *For every admissible $(\tau, \mathbf{H})$ and every $w \in \mathbb{R}^d$,*

$$\frac{1}{T} \sum_{t=1}^T (w - \tau_t)^\top H_t (w - \tau_t) = (w - \bar{\tau}_H)^\top \bar{H} (w - \bar{\tau}_H) + \frac{1}{T} \sum_{t=1}^T (\tau_t - \bar{\tau}_H)^\top H_t (\tau_t - \bar{\tau}_H), \quad (5)$$

and hence $\max_t (w - \tau_t)^\top H_t (w - \tau_t)$ is at least the right-hand side.

The proof is a two-line expansion: the cross-term vanishes by the defining property $\bar{H}\bar{\tau}_H = T^{-1} \sum_t H_t \tau_t$ of the centroid. Two consequences drive everything that follows: components of w orthogonal to $\text{range}(\bar{H}) = V_1 + \dots + V_T$ annihilate against every H_t , so without loss the decoder’s output lies in an at-most- d_{eff} -dimensional subspace; and the bound separates into a compression term in the $\bar{H}$ -metric on $\bar{\tau}_H$ and a floor term that does not depend on w at all.

Lemma 2 (Closed-form floor). *Under P^* with any fixed positive-definite spectra $(D_t)_{t=1}^T$ (possibly distinct across tasks),*

$$\mathbb{E}_{P^*} \left[\frac{1}{T} \sum_{t=1}^T (\tau_t - \bar{\tau}_H)^\top H_t (\tau_t - \bar{\tau}_H) \right] = B^2 \left(1 - \frac{\mathbb{E}[d_{\text{eff}}]}{Tr} \right). \quad (6)$$

The key probabilistic input is the second-moment identity $\mathbb{E}[H_t \tau_t \tau_t^\top H_t \mid U_t] = (B^2/r) H_t$, which holds for any D_t by the $D_t^{-1/2}$ stretch in P^* . Three limits illuminate the formula: at $r = d$ (full rank) the floor is $B^2(1 - 1/T)$, the classical averaged-vector-quantization spread; at $T = 1$ it is zero, recovering single-source quantization; and when all subspaces coincide ($V_t = V$, $d_{\text{eff}} = r$) it is again $B^2(1 - 1/T)$. In the generic LoRA regime (V_t Stiefel-random, $Tr \leq d$), $d_{\text{eff}} = Tr$ almost surely and the floor vanishes.

Theorem 3 (Rank- r RD lower bound, general H_t). *Under P^* with any fixed positive-definite spectra $(D_t)_{t=1}^T$ and $\text{rank}(H_t) = r$ for all t ,*

$$\mathcal{D}^*(T, d, r, B, R) \geq B^2 \left(1 - \frac{\mathbb{E}[d_{\text{eff}}]}{Tr} \right) + c_{\text{TQ}} \cdot \frac{B^2 \mathbb{E}[d_{\text{eff}}]}{Tr} \cdot 2^{-2R/\mathbb{E}[d_{\text{eff}}]}. \quad (7)$$

Under Stiefel-random sampling with $Tr \leq d$, $d_{\text{eff}} = Tr$ a.s., the floor vanishes, and $\mathcal{D}^(T, d, r, B, R) \geq c_{\text{TQ}} B^2 2^{-2R/(Tr)}$, with c_{TQ} the universal constant of TurboQuant (Zandieh et al., 2025) Theorem 3.*

Proof sketch. By (4) it suffices to lower-bound the Bayesian risk under P^* . Lemmas 1 and 2 reduce it to the floor plus the compression term $\mathbb{E}[(w^* - \bar{\tau}_H)^\top \bar{H} (w^* - \bar{\tau}_H)]$; the decoder may be assumed to emit $w^* \in \text{range}(\bar{H})$, and with $\eta := \bar{H}^{1/2} \bar{\tau}_H \in \mathbb{R}^{d_{\text{eff}}}$ the subproblem is rate- R quantization of η under squared loss. Given $\mathbf{H}$, the only remaining randomness is the directions u_t , so a conditional Shannon converse applies to every rate- R code, including codebooks adapted to $\mathbf{H}$: conditioning on $\mathbf{H}$, rather than averaging over rotations, is what makes the converse valid against the $\mathbf{H}$ -aware decoder of §5; the constant absorbs the entropy-power-to-variance ratio of η , which is $O(1)$ because the scale is pinned by $\tau_t^\top H_t \tau_t = B^2$. Combining with $\mathbb{E}[\|\eta\|^2] = B^2 \mathbb{E}[d_{\text{eff}}]/(Tr)$ from Lemma 2 yields (7).

Why the general- H_t statement is the headline. Real LoRA fine-tunes have task-specific curvature: the Fisher information at τ_t is rank- r PSD with range V_t but generally not a projector. Theorem 3 applies with no modeling assumption on the D_t beyond positive-definiteness, closing the MSE $\rightarrow$ CE bridge for the lower-bound half.

5 ACHIEVABILITY: A RATE-MATCHING UPPER BOUND

The lower bound is matched by an explicit algorithm in the generic regime (V_t Stiefel-random, $Tr \leq d$, so $d_{\text{eff}} = Tr$ a.s. and the floor vanishes). The algorithm extends the single-source construction of TurboQuant (Zandieh et al., 2025) with a centroid-reduction step via Lemma 1.

Algorithm. Encoder and decoder share a pseudorandom seed indexing an orthogonal rotation $O \in O(d_{\text{eff}})$ drawn from Haar measure (implemented by QR of an iid Gaussian matrix). *Encoder:* (1) compute $\bar{H}$, $\bar{\tau}_H$, and an orthonormal basis $Q \in \mathbb{R}^{d \times d_{\text{eff}}}$ of $\text{range}(\bar{H})$ with eigenvalues $w_1, \dots, w_{d_{\text{eff}}} > 0$; form $\eta := \bar{H}^{1/2} \bar{\tau}_H$ in the Q -basis. (2) Apply the shared rotation $\hat{\eta} := O\eta_Q$. (3) Uniform-scalar-quantize each coordinate on $[-c\sigma_{\text{pc}}, c\sigma_{\text{pc}}]$ with b bits, where $\sigma_{\text{pc}} := B/\sqrt{Tr}$ and c is a clipping constant (we use $c = 5$); total rate $R = bd_{\text{eff}}$. *Decoder:* invert each step and emit $w^* := \bar{H}^{1/2} \eta_q$ on $\text{range}(\bar{H})$. The centroid $\bar{\tau}_H$ is the sufficient statistic in the floor-zero regime, the $\bar{H}^{1/2}$ scaling whitens it, and the random rotation spreads quantization error equally across directions.

Theorem 4 (Matching achievability, floor-zero Stiefel). *Under P^* with Stiefel-random V_t and $Tr \leq d$, for any fixed positive-definite $(D_t)_{t=1}^T$, the algorithm above at rate R achieves*

$$\mathbb{E}_{P^*} [\max_t (w^* - \tau_t)^\top H_t (w^* - \tau_t)] \leq C \cdot B^2 \cdot 2^{-2R/(Tr)}, \quad C = Tc^2/3. \quad (8)$$

Combined with Theorem 3, the Bayesian rate-distortion function under P^ is $\Theta(B^2 2^{-2R/(Tr)})$ in this regime, universal in d, r, R, B , with the T -dependence of the constant discussed in Remark 5.*

Proof sketch. In the floor-zero regime Lemma 2 forces $H_t \bar{\tau}_H = H_t \tau_t$ almost surely, so worst-task distortion is driven entirely by the quantization error $\delta\eta$ on the whitened centroid; the crude bound $\max_t \leq \sum_t = T \cdot \text{avg}$ gives $\max_t (\delta w)^\top H_t (\delta w) \leq T \|\delta\eta\|^2$, and uniform scalar quantization after the rotation contributes per-coordinate MSE at most $c^2 \sigma_{\text{pc}}^2 2^{-2b}/3$.

Remark 5 (What the Θ hides; the T -dependence is open). The lower-bound constant is T -free while the upper-bound constant is $C = Tc^2/3$; the entire factor T enters at the crude $\max_t \leq T \text{ avg}$ step, which is loose except when one task’s error projection dominates, a case atypical in the floor-zero regime. The synthetic measurements decide this in favour of the benign reading: sweeping $T \in \{2, 4, 8, 16\}$ (1000 trials/cell, Appendix A), the achievability constant grows *logarithmically* ($R^2 \geq 0.998$ for a $\log T$ fit vs. ≈ 0.91 linear; the $T=16$ constant is $1.77\times$ the $T=2$ value, against $8\times$ if linear). The linear T is an artifact of the analysis, not intrinsic; proving the matching $\log T$ (or T -free) constant is open, and the floor-zero Θ should be read as “ Θ up to a $\log T$ factor.”

The shared- V / floor-positive regime. When the V_t overlap ($d_{\text{eff}} < Tr$, floor positive), the zero-floor step fails. A refinement quantizing around the Chebyshev center of the ensemble in an active-set-aware null-space split achieves excess distortion of order $2^{-2R/(d_{\text{eff}}+|A|-1)}$, where $|A|$ is the active-set size: better than naive scalar quantization but short of the lower bound’s $2^{-2R/d_{\text{eff}}}$.

Remark 6 (Shared- V exponent gap). Valid rate- R random-codebook encoders empirically achieve slopes strictly steeper than the linear construction (slope -1.45 vs. the linear prediction -1.20 at $T = 3, r = 3$; Appendix A), so the true rate-distortion function in the floor-positive regime lies strictly between the two bounds. Neither bound is tight; the exact exponent is open.

6 EXPERIMENTS

6.1 SETUP AND SYNTHETIC VALIDATION

Synthetic validation (details in Appendix A). Monte Carlo on the hard distribution P^* across $T \in \{2, 3, 4, 8\}$, $r \in \{4, 8, 16\}$, $d \in \{128, 512, 2048\}$, isotropic and four heavy-anisotropy spectra families: the empirical rate slope is -2.00 ± 0.01 (95% bootstrap CI, 1000 trials/configuration), matching the predicted $-2R/d_{\text{eff}}$ of Theorem 3; the floor $B^2(1 - \mathbb{E}[d_{\text{eff}}]/(Tr))$ is reproduced to four to five digits; and the ratio of measured distortion to the Theorem 3 bound at $c_{\text{TQ}} = 1$ stays ≤ 13 across all configurations, constant in R within bootstrap CI. A dedicated T -sweep (T up to 16) measures the achievability constant’s growth as logarithmic in T (Remark 5). In the floor-positive regime, the null-space construction tracks its predicted exponent $-2r/(r + |A| - 1)$ and a random-codebook encoder is strictly steeper, confirming the open gap of Remark 6.

Real-LLM setup. We calibrate the bound against real LoRA fine-tunes on four instruction-tuned bases spanning distinct pretraining recipes and attention designs: Llama-3.1-8B-Instruct (MHA), Qwen-2.5-7B-Instruct (GQA), Mistral-7B-Instruct-v0.3 (MHA), and Yi-1.5-9B-Chat (MHA). For

each we train $T = 4$ adapters (rank $r = 16$ on the $\{q, k, v, o\}$ projections; AdamW, lr 5×10^{-5} , 7500 examples $\times$ 2 epochs) on GSM8K (math), Alpaca-cleaned (instruction following), Magicoder-OSS-Instruct (code), and WMT19 en $\rightarrow$ de (translation); 16 adapters total, retrained at three independent seeds. For a merged delta w we measure per-task *excess* NLL over the task’s own adapter, $\Delta L_t(w) = L_t(\theta_0 + w) - L_t(\theta_0 + \tau_t)$ on $n = 1000$ held-out examples (nats per response token), and worst-task distortion $\widehat{D}(w) = \max_t \Delta L_t(w)$; the shared frozen base cancels, isolating the cost of merging. CIs are 95% percentile bootstrap over 10,000 resamples. A second cohort, the *T-scaling pool* (the four tasks plus CodeAlpaca, Dolly, XSum), underlies §6.4; excesses are not cross-comparable between cohorts.

6.2 THE REGIME OF REAL MERGING: ZERO FLOOR, 0.10–0.22 NATS OF ALGORITHMIC SLACK

Four findings from the $T = 4$ matrix (five baseline methods: TA, TIES, DARE at density 0.2, KnOTS, and TVQ at $b \in \{1, \dots, 32\}$ bits). **(1) Non-vanishing worst-task excess, zero floor.** At the no-quantization rate, worst-task excess is strictly positive on every base (TA: Llama 0.218, Mistral 0.146, Qwen 0.110, Yi 0.099). Computing $d_{\text{eff}} = \text{rank}(\sum_t P_{V_t})$ directly from the trained adapters gives $d_{\text{eff}} = Tr = 64$ in *every* layer of *every* model: the four task subspaces are linearly independent, the predicted floor is exactly zero, and the 0.10–0.22 nats gap is method suboptimality, not an unavoidable cost of merging. Deliberate attempts to break the saturation (90% shared training data, rank $r = 64$) fail to move it; a mechanical barrier explains why ($d_{\text{eff}} < Tr$ requires $r > d_{\text{in}}/T$, far outside the practical LoRA range; Appendix B, with a floor-estimation recipe). **(2) Merging difficulty tracks training recipe, not architecture.** The excess orders Llama $>$ Mistral $>$ Qwen $>$ Yi (non-overlapping CIs), cutting across attention design and unexplained by either hard rank or a soft participation-ratio overlap statistic. **(3) Only TIES separates from the structure-blind cluster.** TIES cuts worst-task excess by 30–88% vs. TA (3-seed means), concentrated on the bottleneck task; KnOTS and DARE are statistically indistinguishable from TA (paired differences ≤ 0.003 nats), exactly as the theory predicts at zero subspace overlap, where alignment has nothing to exploit. **(4) A universal “less-is-more” dip at $b = 2$.** The TVQ rate sweep is non-monotone on every base: 2-bit quantization sits $1.7\text{--}6.0\times$ below the adjacent $b = 4$ point (non-overlapping CIs), while $b = 1$ and $b \geq 4$ regress to the $\approx$ TA floor; the dip covaries with TIES per-task but is *not* explained by implicit sparsity (Appendix C).

Seed robustness. Retraining the matrix at three independent seeds and running 10,000-resample paired bootstraps on the five method-pair gaps behind our recommendations leaves *all* 20 gaps *Confident* (largest CI half-width ± 0.0044 nats; Appendix H). The $d_{\text{eff}} = Tr$ finding is empirical (trained subspaces are optimized, not random) but holds in 100% of layers across four independently pretrained models.

6.3 THE EXACT ENCODER FAILS ON REAL ADAPTERS; ONE RIDGE TERM SALVAGES IT

We instantiate the encoder of Theorem 4 with the projector surrogate $H_t = P_{V_t}$ (top- r right-singular projector of each task’s delta), evaluated at the $b \rightarrow \infty$ rate on Llama-3.1-8B at $T = 4$. **The exact centroid fails:** worst-task excess 0.340 (3-seed mean), *worse* than TA (0.220) and more than $2\times$ TIES (0.154). The mechanism is spectral: $\bar{H}$ ’s non-zero spectrum is nearly degenerate (median $\approx 2 \times 10^{-3}$, minimum $\approx 10^{-5}$), and the pseudo-inverse divides by these sliver eigenvalues, amplifying $\|w^*\|$ to $25\text{--}94\times$ the TA norm, far outside the quadratic surrogate’s validity region; the failure is in the surrogate $\rightarrow$ CE bridge, not in the construction’s surrogate optimality.

The salvage. Replacing $\bar{H}^+$ with $(\bar{H} + \lambda I)^{-1}$, the same construction with a numerical sliver-floor exposed as one hyperparameter, produces a clean U-curve with minimum at $\lambda^* = 0.05$: worst-task excess **0.094**, 57% below TA and 39% below TIES (Figure 2). The sweep, per-task breakdown, and a Fisher-diagonal sanity check are in Appendix G. The salvage generalizes: each base has a well-defined optimum ($\lambda^* = 0.05$ on Llama, 0.13 on the other three), and the *single globally-fixed* $\lambda = 0.13$ (the leave-one-base-out modal choice) already beats TA and TIES on all four bases (0.125/0.044/0.010/0.037).

Head-to-head and the tuning question. Table 1 places rd-encoder ridge in one seed-consistent comparison with nine baselines, including its closest structural relative, data-free RegMean (Jin et al., 2023), and data-free AdaMerging (Yang et al., 2024): lowest worst-task excess on every base

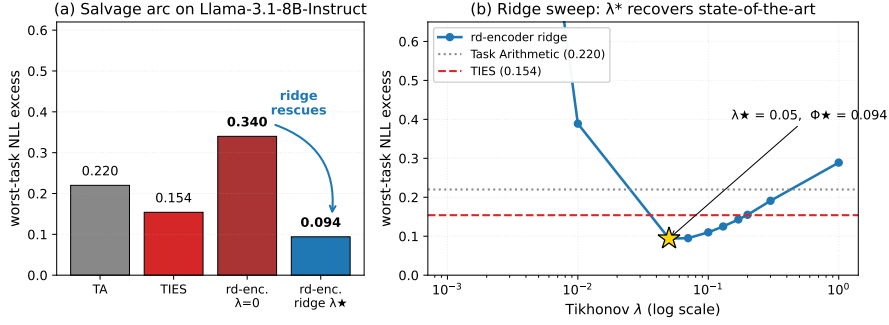


Figure 2: **Salvage arc.** (a) On Llama-3.1-8B-Instruct at $T = 4$, the exact rd-encoder construction ($\lambda = 0$) collapses to worst-task excess 0.340, above TA (0.220) and TIES (0.154); Tikhonov regularization at $\lambda^* = 0.05$ recovers 0.094, the lowest figure for this base. (b) rd-encoder ridge sits below both TA and TIES across the band $[0.03, 0.18]$. All values are 3-seed means on matched seed-1/2/3 adapters.

Table 1: Unified head-to-head: worst-task NLL excess ($T = 4$ matrix) for ten methods at default hyperparameters. RD-ENCODER RIDGE (base-optimal λ) is its 3-seed mean on matched seed-1/2/3 adapters; the four added closed-form baselines are seed-1 cells (seed-stable to ± 0.01 ; Appendix H).

Method	Llama-3.1	Mistral-7B	Qwen-2.5	Yi-1.5
Task Arithmetic	0.213	0.133	0.104	0.099
TIES	0.147	0.049	0.013	0.045
DARE	0.213	0.134	0.107	0.101
KnOTS	0.213	0.132	0.104	0.099
TVQ ($b=2$)	0.101	0.054	0.019	0.057
Fisher-magnitude	0.148	0.050	0.014	0.053
DELLA	0.131	0.052	0.014	0.055
RegMean (data-free)	0.243	0.054	0.034	0.045
AdaMerging (data-free)	0.228	0.113	0.087	0.075
RD-ENCODER RIDGE	0.094	0.044	0.010	0.037

at default hyperparameters. Because a swept λ against untuned baselines would be an unfair win, we also sweep each close competitor’s key hyperparameter (TIES/DARE density over five values, RegMean’s own ridge λ over six decades; Appendix G): rd-encoder ridge remains best on Llama, Mistral, and Qwen against every tuned competitor, but on saturated Yi-Chat a tuned RegMean edges it (0.033 vs. 0.037, consistent across three matched seeds) and a tuned TIES ties it. We therefore claim the strongest data-free merge *robustly on the non-saturated bases* and competitive-but-not-dominant on saturated Yi, the base where we independently flag NLL excess as an unreliable proxy (R2). Part of the default-hyperparameter Llama gap over RegMean (0.243) is the same sliver pathology that breaks our own $\lambda = 0$ construction (0.340): RegMean has no ridge term, which is precisely why the tuned-RegMean control matters.

6.4 T -SCALING AND THE TIES-INVERSION AUDIT

To test whether the ordering survives beyond $T = 4$, we sweep $T \in \{2, 4, 7\}$ on three bases (Yi-1.5-9B-Chat, Llama-3.1-8B-Instruct, and pre-registered Mistral-7B-v0.3) over the seven-task pool (162 merge cells, six methods, identical subset composition across bases). Figure 1 plots the sweep; the full numbers are in Table 6 (Appendix D).

The salvage grows with T . On Llama, rd-encoder ridge is best at every T with a widening margin (ratio rd-ridge/TA $0.84 \rightarrow 0.68 \rightarrow 0.52$); the same strengthening holds on Mistral ($1.38 \rightarrow 0.82 \rightarrow 0.47$, computed from unrounded excess): rd-ridge loses narrowly at $T = 2$ but wins outright at $T = 7$. Worst-task excess follows $a + b \log T$ on all three bases ($R^2 \geq 0.93$ on 16 of 18 fits), the real-data counterpart of the synthetic $\log T$ prediction, and rd-ridge has the lowest and most base-invariant slope (0.070/0.065/0.062 on Yi/Llama/Mistral vs. 0.079–0.171 for the matrix methods).

Table 2: Downstream accuracy (3-seed means). Spearman ρ is computed across the five matrix baselines between accuracy and $-NLL$ excess: positive ρ means NLL ranking predicts accuracy ranking. Each cell has $n = 5$ methods and is individually low-powered; the signal is the across-cell pattern, 7 of 8 cells positive. Falsification tests and per-seed stability in Appendix E.

	TA	TIES	DARE	KnOTS	TVQ ₂	rd-ridge	ρ
<i>GSM8K exact-match</i>							
Llama-3.1-8B	0.315	0.256	0.317	0.323	0.299	0.201	−0.60
Mistral-7B-v0.3	0.103	0.409	0.103	0.097	0.379	0.469	+0.67
Qwen-2.5-7B	0.223	0.706	0.208	0.218	0.685	0.794	+1.00
Yi-1.5-9B	0.763	0.783	0.751	0.759	0.769	0.791	+0.90
<i>HumanEval pass@1</i>							
Llama-3.1-8B	0.055	0.354	0.059	0.057	0.441	0.457	+1.00
Mistral-7B-v0.3	0.053	0.250	0.057	0.045	0.244	0.191	+0.60
Qwen-2.5-7B	0.016	0.547	0.014	0.016	0.644	0.606	+0.87
Yi-1.5-9B	0.061	0.081	0.069	0.069	0.081	0.108	+0.72

The TIES inversion, and an audit that predicts it. On Yi, TIES becomes the *worst* method at $T = 7$; on Llama and Mistral it does not. A per-coordinate sign-vote probe on the $T = 7$ cohorts (CPU-only, a few minutes per cohort) locates the difference not in vote-split structure but in the *per-task win-share range* R : 0.077 on Yi vs. 0.028 on Llama (Appendix D): on a saturated base tasks sharing instruction-following structure dominate the vote, and the code and near-saturated adapters are systematically outvoted. A perturbation model of the TIES update (Appendix I) converts this into a threshold: predicted ambiguity window $R^* \in [0.025, 0.075]$, with inversion above it and TIES winning outright below it. Yi ($R = 0.077$, above) inverts; Llama ($R = 0.028$, inside) is ambiguous. We then **pre-registered** an out-of-sample test (commit 3582799 in the supplementary audit trail, timestamped before the Mistral sweep ran): Mistral-7B, probe range $R = 0.0326$, inside the window, predicted ambiguous at $T = 7$ under operationalized criteria fixed in advance (“clearly worst” = last and $> TA + 0.04$; “clearly best” = first by > 0.02). The sweep confirms it: TIES second of six, 0.092 nats *below* TA, so neither condition fires. A base-free synthetic experiment closes the causal loop: a single sign-conflict knob raises R and the TIES–TA gap in lockstep (Pearson 0.998; Appendix I). *Caveats, stated plainly*: the window’s constants were fit to the two in-sample points, “ambiguous” is the prior-likely outcome for most methods, and the below-window prediction is untested (no cohort we measured has $R < 0.028$). We present the threshold as a calibrated heuristic with one out-of-sample consistency check, not a validated law.

6.5 DO NLL CONCLUSIONS HOLD ON TASK ACCURACY?

NLL excess is fast and seed-stable but not what a practitioner deploys against, so we re-merge the $T = 4$ matrix and evaluate GSM8K exact-match ($n = 500$, greedy CoT) and HumanEval pass@1 ($n = 164$, sandboxed) on matched seed-1/2/3 adapters (3-seed means; Table 2); rd-encoder ridge runs at the λ chosen on NLL and never sees an accuracy metric.

Headline. The NLL→accuracy correspondence is positive on 7 of 8 base×metric cells, meeting the pre-specified $\rho \geq 0.7$ threshold on 5; the single inversion (Llama-3.1, GSM8K EM, $\rho = -0.60$) does not replicate on HumanEval, and both pre-specified mechanism candidates for it were falsified (Appendix E). The methods that minimize NLL excess (TIES, TVQ₂, rd-ridge) also produce far more deployable code merges: TA/DARE/KnOTS cluster 2–40× below them on pass@1.

RD-ENCODER RIDGE on accuracy: strong, not a sweep. rd-ridge is best of six on *five of eight* cells; its GSM8K wins on Qwen (+0.088 over TIES) and Mistral (+0.060) are an order of magnitude larger than its across-seed SD there. On two of the three cells it does not lead, Llama-3.1 GSM8K (worst of six, 0.201) and Mistral HumanEval (third, 0.191), it is *seed-unstable* (across-seed SD 0.074 and 0.084 vs. ≤ 0.015 on the cells it wins or ties): the norm-amplifying ridge construction can push greedy generation into degenerate regions on particular adapter draws, a fragility invisible to the seed-stable NLL it was selected on. An earlier draft claimed best-or-tied on all eight cells from a single-draw evaluation; on matched three-seed means that claim is **retracted**. The robust result is the seed-for-seed NLL worst-task-excess win on all four bases; the deployment reading is to select by NLL and verify on ≥ 2 downstream metrics.

6.6 PRACTICAL RECOMMENDATIONS

Five rules distilled from §6.2–§6.5 (decision table in Appendix J). **R1: Default to rd-encoder ridge.** Lowest worst-task NLL excess on every base at $T = 4$, strictly best at $T = 7$ on Llama and Mistral; without any sweep, the global $\lambda = 0.13$ still beats TA and TIES on all four bases. The robust win is on NLL; accuracy is a majority-of-cells result (§6.5). **R2: Audit base saturation before trusting NLL comparisons.** On heavily chat-tuned bases, compressed per-task headroom amplifies excess asymmetries that need not track deployable quality (Yi’s near-saturated Alpaca adapter; Appendix D); renormalize by headroom or exclude near-saturated adapters; a TIES win-share range above ~ 0.05 is the warning signal. **R3: If not rd-encoder ridge, use TIES or TVQ- $b=2$, not TA; avoid TIES on saturated bases.** TIES wins 3 of 4 bases at $T = 4$ but inverts on Yi at $T = 7$; the win-share audit of §6.4 predicts the regime. **R4: Verify on ≥ 2 downstream metrics before deploying.** NLL predicts accuracy on 7 of 8 cells, but the L3-GSM8K cell shows single-metric verification can mislead. **R5: Estimate the floor before assuming you are algorithm-bound.** The recipe of Appendix B computes $\Phi_{\text{lower}}^* = B^2(1 - d_{\text{eff}}/(Tr))$ from adapter artifacts in $\lesssim 1$ CPU-minute per task; every cohort we measured is floor-zero.

7 DISCUSSION AND LIMITATIONS

Cross-entropy extension. Theorem 3 is proved for quadratic loss with H_t rank- r PSD; for LLM cross-entropy the natural H_t is the Fisher information at τ_t , and the proof uses only $H_t H_t^+ H_t = H_t$ and the $O(d)$ -equivariance of the hard distribution, both of which survive verbatim. A fully rigorous MSE-to-CE bridge must control the Taylor remainder across the H_t -ball; the rate budget makes the remainder third-order in the quantity the quadratic term is second-order in, and $M_t \leq \text{poly}(L)$ for softmax-CE with bounded logits. We state this as an informal, remainder-bounded estimate (sketch in the supplementary material); the empirical sections measure CE excess directly, and the E11 probe (Appendix F) confirms rd-ridge approaches the Fisher-quadratic diagonal optimum to within $\sim 20\%$ in a base-dependent α window.

Scope of the hard distribution. The lower bound is established against P^* ; real fine-tunes exhibit structure iid Stiefel sampling does not capture (semantically correlated subspaces, anisotropic curvature), so the theorems characterize the floor for merging, not a tight prediction of any specific real-LLM experiment. What the real-LLM study adds is the measured regime: floor zero in every layer of every base, so the 0.10–0.22 nats of excess is distance from the optimum; the cross-model ordering of that slack is unexplained by subspace geometry, pointing to the loss curvature H_t and the training recipe as the operative quantities.

Known limitations (in descending order of constraint). (1) **Single LoRA rank:** all experiments use $r = 16$; the $\rho \approx 1/\sqrt{r}$ slope prediction is consistent with but not verified beyond it. (2) **Four bases:** Llama-3.1, Mistral-7B, Qwen-2.5, Yi-1.5 span an instructive range of recipes and saturation regimes but exclude, e.g., Gemma and Phi. (3) **Uneven seed coverage:** the $T = 4$ matrix has three training seeds (all 20 method-pair gaps Confident); the T -scaling sweep and added baselines use one. (4) **rd-encoder ridge downstream accuracy is strong but not uniform:** best on five of eight cells, seed-unstable and not best on two (§6.5); R1 rests on the NLL win. (5) **One unexplained outlier cell:** the (Llama-3.1, GSM8K EM) inversion survives both pre-specified mechanism tests (Appendix E). (6) **The quadratic-bridge window is base-dependent** (small- α on Yi, $\alpha \in [0.15, 0.25]$ on Llama; Appendix F). (7) **The Td2 threshold is order-of-magnitude:** its constants are estimated by spectral analogy, not first-principles Fisher information, and the below-window edge is untested (Appendix I). (8) **Not an exhaustive method survey:** DARE-TIES and the newest 2026-era methods are discussed but not re-run; the headline is robust to the absence of any single further baseline.

Closing. We have given LoRA merging the information-theoretic reference point it was missing: worst-task NLL excess splits into a task-overlap floor and a compression term, an explicit encoder closes the gap up to a logarithmic constant in T , and on every real cohort we tested the floor is zero: what current methods lose is recoverable slack, a measurable fraction of which our regularized encoder claims. Merging research can now be judged not only against the next benchmark, but against a measurable theoretical ceiling.

REPRODUCIBILITY STATEMENT

All theoretical claims are stated with explicit assumptions in §3 and proved in §4–§5 and the supplementary material. For the empirical results we release, on publication, the training and evaluation code, exact per-task configuration files (base model, LoRA rank and target modules, optimizer, token budget, eval split sizes), random seeds, and the per-example NLL arrays underlying every cell of the merge matrix. Training and evaluation splits are disjoint (verified by a separate zero-overlap audit); confidence intervals are 95% percentile bootstrap over 10,000 resamples, with paired bootstraps for method-to-method comparisons; point estimates were reproduced across two independent evaluation runs to within 0.001 nats/token. The headline $T = 4$ matrix uses three independent training seeds (Appendix H); the T -scaling sweep and added baselines use a single training seed, as stated in their sections. The pre-registered Mistral prediction of §6.4 is verifiable from the anonymized, timestamped experiment-log bundle in the supplementary material, whose commit trail shows no result commit predating the registration. Appendix J gives a per-experiment manifest (cells, reproducer script, per-cell wallclock); the total footprint of the empirical results is roughly 80–120 GPU-hours on a single A100-80GB plus adapter training. A chronological log of design decisions and intermediate negative results accompanies the camera-ready version.

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A SYNTHETIC VALIDATION DETAILS

All runs use the hard distribution P^* of §3 with task-dependent spectra $(D_t)_{t=1}^T$. Code, random seeds, and raw outputs are released with the paper.

Lower-bound slope. We sample ensembles across $T \in \{2, 3, 4, 8\}$, $r \in \{4, 8, 16\}$, $d \in \{128, 512, 2048\}$, with both isotropic spectra ($D_t = I_r$) and four heavy-anisotropy families (geometric, linear, two-scale, and mixed; spectra spread across $[2^{-1}, 2^1]$). For each configuration we measure the excess $\mathbb{E}_{P^*}[\|\bar{H}^{1/2}w^* - \eta\|^2]$ of the Gaussian-QR + uniform scalar quantization algorithm of §5 as a function of $R = bd_{\text{eff}}$ bits, and fit a linear regression of $\log_2 \mathbb{E}[\text{excess}]$ against $b = R/d_{\text{eff}}$. Across all tested configurations the empirical slope is -2.00 ± 0.01 (95% bootstrap CI across 1000 trials per configuration), matching the prediction $-2R/d_{\text{eff}}$ of Theorem 3. The floor term $B^2(1 - \mathbb{E}[d_{\text{eff}}]/(Tr))$ is reproduced to four to five digits of precision in the common- D_0 and task-dependent- D_t regimes.

Achievability constant and its T -dependence. We measure the ratio $\mathbb{E}_{P^*}[\max_t D_t(w^*)]/(c_{\text{TQ}} B^2 2^{-2R/(Tr)})$ of empirical upper bound to the Stiefel lower bound at $c_{\text{TQ}} = 1$. The ratio is constant in R (a necessary condition for the upper bound to match the lower bound in order) and starts near 11 at $T = 2$ ($r = 4$, $d = 128$), consistent with $C = Tc^2/3 \approx 17$ at $c = 5$ combined with $c_{\text{TQ}} \approx 1$ from TurboQuant (Zandieh et al., 2025). Sweeping $T \in \{2, 4, 8, 16\}$ (1000 trials/cell) settles the T -dependence flagged in Remark 5: the constant grows *logarithmically*, not linearly; a $\log T$ fit gives $R^2 \geq 0.998$ against $R^2 \approx 0.91$ for a linear fit, and the $T=16$ value is only $1.77\times$ the $T=2$ value, far below the $8\times$ a strictly linear- T factor would give. The linear T in $C = Tc^2/3$ is thus an artifact of the crude $\max_t \leq T$ avg step, not intrinsic to the rate-distortion function; the matching $\log T$ achievability constant is stated as a conjecture.

Shared- V regime. For the floor-positive regime we implement the null-space-aware bit allocation of §5 with per-coordinate fractional bits and a locked clip $c = 11.5\sigma_{\text{pc}}$ (tuned once, applied across all configurations). At $T = 2$, $r = 4$, across five anisotropy mixes (iso+iso, iso+geom, geom+twoscale, lin+twoscale, geom+iso), the empirical slope of excess over cheb² is -1.60 ± 0.10 at $n_{\text{trials}} = 1000$, matching the theoretical prediction $-2r/(r + |A| - 1) = -8/5$ with bootstrap 95% CI ± 0.010 per cell. The ± 0.10 spread reflects structural anisotropy: iso+iso tracks closer to the scalar-quantization limit -2 ; geom+twoscale undershoots due to H_t -metric distortion on the perpendicular subspace. Extending to $T = 3, 4$ via a SOCP Chebyshev solver with Gauss–Newton KKT refinement (residual $\sim 10^{-15}$), the null-space split continues to track the theoretical prediction across mixed anisotropies.

Random-codebook comparison. As a sanity check on the tightness of the linear construction in the shared- V regime, we ran a valid rate- R random-codebook encoder (2^R iid Gaussian codewords, encoder picks the one minimizing $\max_t D_t$, $R \leq 18$). At $T = 3$, $r = 3$, the random-codebook slope is -1.45 , strictly steeper than the explicit linear construction’s -1.20 . This rules out sharpening the lower bound to match the linear upper-bound exponent; the true rate-distortion function in the floor-positive regime lies strictly between $-2r/r = -2$ and $-2r/(r + |A| - 1)$ and is left open (Remark 6).

Table 3: Subspace metrics at Qwen-2.5-7B’s 112 attention projections under the two arms. Neither breaks the $d_{\text{eff}}/(Tr) = 1$ saturation in any majority sense.

Condition	hard mean	hard min	soft mean	layers < 0.8
Mixture, $\alpha = 0.0$	1.000	1.000	0.257	0/112
Mixture, $\alpha = 0.5$	1.000	1.000	0.253	0/112
Mixture, $\alpha = 0.9$	1.000	1.000	0.251	0/112
Forcing, $r = 64$	0.997	0.867	0.253	0/112

B THE FLOOR-POSITIVE REGIME DOES NOT ARISE NATURALLY

B.1 WHY THE FLOOR FORMULA IS OPERATIONALLY VACUOUS ON REAL LoRA FINE-TUNES

On the sixteen adapters of §6.2 we measured $d_{\text{eff}} = Tr$ in every attention projection of every base model, so the predicted floor $B^2(1 - d_{\text{eff}}/(Tr))$ is identically zero. A natural concern: is this an unlucky choice of task corpora or hyperparameters? We tested two operational routes to break the saturation.

Arm 1: data-mixture overlap. On Qwen-2.5-7B we trained 4 task adapters under three data-mixture levels: at mixing parameter $\alpha \in \{0, 0.5, 0.9\}$ the training data for the target task is a fraction α from a fixed shared pool (uniformly sampled across all four task corpora) and $(1 - \alpha)$ from the target task’s own set. At $\alpha = 0.9$ four adapters share 90% of their training examples. Per-layer hard $d_{\text{eff}}/(Tr)$ stayed saturated at 1.000 across all 112 attention projections at every α (Table 3); the soft (participation-ratio) statistic drifted by 2%, inside seed noise. *Real fine-tuning resists subspace overlap*: the top- r right-singular basis of Δ_t is set by the per-task answer-distribution loss, not the inputs’ raw geometry, so shared training inputs do not induce shared LoRA subspaces.

Arm 2: geometric forcing at $r = 64$. On the same base we trained the four adapters at rank $r = 64$ ($Tr = 256$). Hard $d_{\text{eff}}/(Tr)$: mean 0.997, min 0.867, 0/112 layers below the pre-specified 0.8 threshold. The floor formula still evaluates to zero in the majority of layers and is operationally negligible across the network.

The mechanical-forcing barrier. Generically $d_{\text{eff}} = \min(d_{\text{in}}, Tr)$, so $d_{\text{eff}}/(Tr) < 1$ requires $r > d_{\text{in}}/T$: at $T = 4$ this is $r > 1024$ on Llama-3.1-8B, $r > 896$ on Qwen-2.5-7B, $r > 768$ on Llama-3.2-3B, all far outside the practical LoRA regime ($r \leq 64$ in nearly all production deployments). Mechanically forcing the floor-positive regime would require LoRAs of rank comparable to the hidden dimension, defeating the parameter-efficiency that motivates LoRA.

Interpretation. The lower bound continues to apply (it is an information-theoretic floor that always holds) but its non-vacuous regime sits outside the practical LoRA setting in any natural way we could probe. This is a *positive* finding for the achievability program: the algorithmic regime, where worst-task excess is dominated by construction-side suboptimality rather than the irreducible floor, is the regime all real LoRA fine-tunes live in. We do not rule out a deliberately engineered floor-positive regime (shared adapter modules, hard-coded subspace constraints, extreme high-rank LoRA); constructing one is future work.

B.2 RECIPE: ESTIMATING THE FLOOR ON TRAINED ADAPTERS

The recipe below is what we used above and what practitioners can re-run on their own checkpoints; it needs only the adapter artifacts, not the base model.

Step 1: per-task, per-layer projector. The effective dimension is fundamentally a per-layer quantity. For each attention projection in each layer ℓ , form the adapter’s column matrix $M_t^\ell \in \mathbb{R}^{d_{\text{in}} \times r_t}$ and its rank- r_t orthonormal projector $V_t^\ell = \text{qr}(M_t^\ell).Q$. (A single global stack across layers would force $d_{\text{eff}} = Tr$ trivially and obscure the per-layer geometry the barrier depends on.)

Table 4: Worst-task NLL excess of explicit magnitude-pruning at the densities matching $b = 2$ TVQ’s implicit sparsity, with TVQ $b = 2$ and $b = 4$ for reference.

Model	magprune $\rho = 0.50$	magprune $\rho = 0.13$	TVQ $b = 2$	TVQ $b = 4$
Llama-3.1-8B	0.225	0.288	0.108	0.217
Mistral-7B-v0.3	0.155	0.251	0.058	0.138
Qwen-2.5-7B	0.118	0.229	0.019	0.105
Yi-1.5-9B	0.088	0.151	0.059	0.098

Step 2: stacked projector and SVD. Per layer, concatenate $\mathbf{V}^\ell = [V_1^\ell, \dots, V_T^\ell]$ and take its singular values $\{\sigma_i\}$. The *hard* per-layer effective dimension is $d_{\text{eff}}^{\text{hard}, \ell} = |\{i : \sigma_i > 10^{-3}\sigma_1\}|$; the *soft* version is the participation ratio $(\sum_i \sigma_i^2)^2 / \sum_i \sigma_i^4$.

Step 3: per-coordinate radius B . For TVQ at b bits, B is the 99.9th percentile of $|\Delta_{t,i}|$ across tasks (the strict max is dominated by 1–2 extreme coordinates); for continuous methods, the same quantile of $|\Delta_t|$ summed across tasks.

Step 4: floor and algorithmic gap. $\Phi_{\text{lower}}^* = B^2(1 - d_{\text{eff}}/(Tr))$, evaluated per layer (the Lemma 2 form). If $d_{\text{eff}}^\ell = Tr$ everywhere (the regime all our measurements occupy), the floor is identically zero and any measured worst-task excess is the *entire* algorithmic gap. If $d_{\text{eff}} < Tr$, the floor bounds the smallest excess any encoder at any rate can achieve, and only $\Phi_{\text{meas}}^* - \Phi_{\text{lower}}^*$ is recoverable.

Caveats. The adapters must share a base model; the soft statistic is sensitive to layer-norm scaling (normalize each layer’s Δ_t to unit Frobenius norm when the scale varies); and the recipe is a diagnostic floor only; it does not predict downstream accuracy. A single pass over the adapter directories costs $\lesssim 1$ CPU-minute per task at $r = 16$.

C WHAT DRIVES THE $b = 2$ DIP

The $T = 4$ matrix (§6.2) shows that 2-bit task-vector quantization often outperforms higher bit budgets at worst-task NLL excess, the “less-is-more” dip that survives on all four bases. An *implicit-TIES* explanation is natural: $b = 2$ quantization implicitly induces sparsity, playing the same role as TIES’s explicit magnitude trimming. We pre-registered two falsifiable predictions of that hypothesis and tested both.

Prediction 1 (covariance with TIES): CONFIRMED. If $b = 2$ TVQ and TIES share a mechanism, their per-task excess patterns should covary across the matrix. Pairing each (model, task, seed) cell’s dip $_t := \Delta L_t^{(b=4)} - \Delta L_t^{(b=2)}$ with its ties $_{\text{win}_t} := \Delta L_t^{(\text{TA})} - \Delta L_t^{(\text{TIES})}$, the Spearman correlation across the 32 cells is $\rho = +0.96$ (Pearson $+0.95$), above the pre-registered 0.7 threshold; per-model it holds 4/4 (Llama 0.98, Mistral 0.95, Qwen 0.95, Yi 0.81), per-task 3/4 (GSM8K is a within-task null where both interventions reliably help but the variation does not covary).

Prediction 2 (sparsity is the shared mechanism): REJECTED. $b = 2$ quantization does induce substantial implicit sparsity: 50% of coordinates land in the zero bucket, and quantization error exceeds the original magnitude for $\sim 87\%$ of coordinates. To test whether sparsity drives the dip, we built an explicit magnitude-pruning merge (keep top- ρ by $|\Delta|$, zero the rest, sum, SVD-truncate to rank r) at the two densities matching those measurements. Pruning at matched sparsity does *not* reproduce the dip; in 7 of 8 cells it underperforms even $b = 4$ (Table 4).

Sharpened hypothesis. The covariance with TIES is real but not mediated by implicit sparsity. Two candidates remain open: (i) *uniform shrinkage of large coefficients*: a min-max $b = 2$ quantizer pulls the largest coefficients toward the surviving bucket centers, and TIES’s sign election undercounts sign-disagreeing coefficients, so both compress the merged-coefficient distribution; (ii) *per-tensor (rather than per-coordinate) calibration*: both methods re-anchor at the tensor level, which global-density magnitude pruning lacks. The present finding: the $b = 2$ dip is real, covaries with

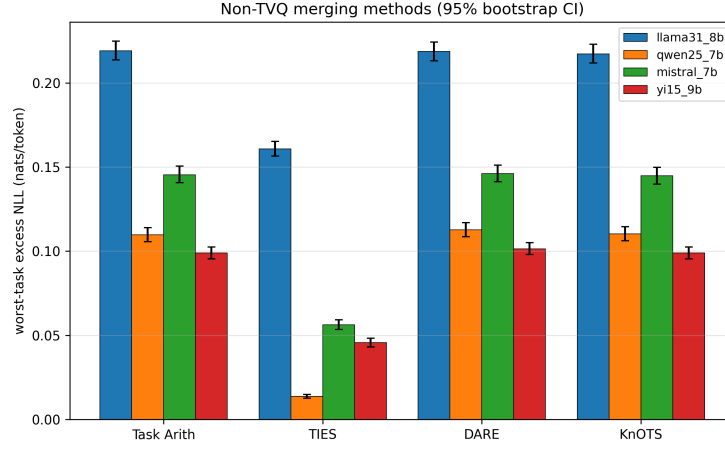


Figure 3: Non-TVQ merging methods, four architectures, 95% bootstrap CIs. Only TIES separates from the Task-Arithmetic / DARE / KnOTS cluster.

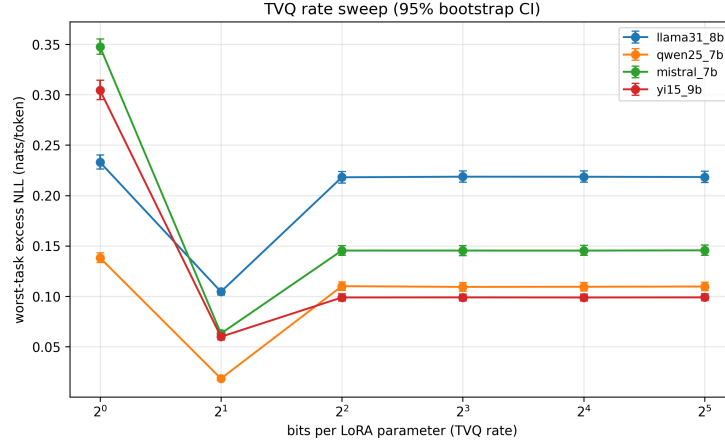


Figure 4: TVQ rate sweep, four architectures, 95% bootstrap CIs. A universal local minimum at $b = 2$ sits well below the adjacent rates on every model; $b = 1$ (too coarse) and $b \geq 4$ (too fine) regress to the $\approx$ Task-Arithmetic floor.

TIES per-task, and is *not* explained by implicit sparsity; $b = 2$ and TIES behave alike because both impose tensor-level structure on the merge, not because $b = 2$ trims small coefficients.

D T -SCALING: PROBE DETAILS AND BASE-SPECIFIC ARTIFACTS

This appendix collects the T -scaling details behind §6.4: the sweep setup, the log- T slope fits, the TIES sign-election probe, and the three Yi-specific artifacts the cross-model sweep resolves.

Setup. For each base we trained three additional rank-16 adapters (CodeAlpaca, Dolly, XSum) and merged them with the four $T = 4$ -matrix adapters. Six methods span the matrix: TA, TIES, DARE, KnOTS, TVQ₂, and rd-encoder ridge at the base-optimal λ . For each T we draw one fixed nested chain plus three random subsets at $T \in \{2, 4\}$ and the all-7-tasks subset at $T = 7$; each cell evaluates worst-task NLL excess at $n_{\text{eval}} = 1000$. Total: 162 merge cells (54 per base, identical subset composition across bases).

Log- T slopes. Table 5 fits $\text{worst-excess}(T) = a + b \log T$ per (base, method). The synthetic estimate (§6.1) gives $b \approx 1/\sqrt{r} = 0.25$ at $r = 16$; measured slopes are positive with the functional

form holding at $R^2 \geq 0.93$ on 16 of 18 all-cells fits (the exceptions are the Yi and Mistral TVQ₂ fits), with the empirical ρ between half and two-thirds of the synthetic estimate on Llama/Mistral. Rd-ridge has the lowest and most base-invariant slope.

Table 5: Log- T slope b in worst-excess = $a + b \log T$ per (base, method). Yi alpaca-OUT uses $T \in \{2, 4\}$ only (no $T = 7$ subset is Alpaca-free).

base	split	TA	TIES	DARE	KnOTS	TVQ ₂	rd-ridge
Yi	All	0.079	0.098	0.082	0.079	-0.001	0.070
Yi	alpaca-OUT	0.046	0.053	0.047	0.046	-0.088	0.031
Llama	All	0.156	0.133	0.155	0.156	0.119	0.065
Mistral	All	0.171	0.095	0.171	0.171	0.095	0.062

Full worst-task excess table. Table 6 gives the numbers behind Figure 1, including the Alpaca-excluded Yi splits (no $T = 7$ subset is Alpaca-free).

Table 6: Worst-task NLL excess (mean across n subsets) per (base, T , method), T -scaling pool. Yi all-cells include the Alpaca adapter, which is near-saturated on Yi-Chat (headroom 0.114 vs. 0.236–0.554 for the others); alpaca-OUT excludes Alpaca-containing subsets. Llama has no shallow-adapter artifact.

base	split	T	TA	TIES	DARE	KnOTS	TVQ ₂	rd-ridge
Yi	All	2	0.027	0.025	0.027	0.027	0.111	0.028
Yi	All	4	0.067	0.095	0.067	0.066	0.101	0.084
Yi	All	7	0.128	0.148	0.130	0.128	0.111	0.115
Yi	alpaca-OUT	2	0.024	0.021	0.023	0.024	0.126	0.023
Yi	alpaca-OUT	4	0.056	0.058	0.056	0.056	0.064	0.045
Llama	All	2	0.068	0.064	0.069	0.068	0.064	0.057
Llama	All	4	0.153	0.126	0.153	0.153	0.113	0.104
Llama	All	7	0.265	0.232	0.265	0.265	0.215	0.138
Mistral	All	2	0.027	0.032	0.027	0.027	0.055	0.038
Mistral	All	4	0.108	0.095	0.108	0.108	0.082	0.088
Mistral	All	7	0.244	0.152	0.244	0.244	0.177	0.115

The TIES sign-election probe. To test the conjecture that sign-election consensus degrades faster on Yi than Llama, we ran a per-coordinate sign-vote audit on both $T = 7$ cohorts: trim each adapter’s Δ at density 0.2 (the TIES default), stack the trimmed deltas, and tally per-coordinate votes plus per-task win-share (fraction of coordinates where that task’s sign matches the elected sign). Table 7 reports the result: vote-split structure is nearly identical between the bases (rejecting the literal “consensus degrades” conjecture) but the *per-task win-share range* is $2.8\times$ wider on Yi. The refined mechanism is a *directionally biased* consensus, not a degraded one: on Yi-Chat, six adapters sharing instruction-following structure vote in a common subspace, and the code adapter (Magicoder, win share 0.776) and the near-saturated Alpaca adapter (0.785) are systematically outvoted; on Llama the per-task votes are nearly uniform (0.805–0.832). The probe is CPU-only and takes a few minutes per cohort.

Table 7: TIES sign-election audit on the $T = 7$ cohorts. Win-share ranges are computed from unrounded shares (Llama: $0.8323 - 0.8045 = 0.0278 \rightarrow 0.028$, above the $R_{lo}^* = 0.025$ edge).

metric	Yi-1.5-9B-Chat	Llama-3.1-8B-Instruct
mean vote margin	0.748	0.753
unanimous coords	69.75%	70.43%
thin-split (± 1)	27.27%	26.80%
wide-split (> 1)	2.98%	2.77%
per-task win-share range	0.077	0.028
win-share min (task)	0.776 (Magicoder)	0.805 (GSM8K)
win-share max (task)	0.853 (GSM8K)	0.832 (CodeAlpaca)

Three Yi-specific artifacts the cross-model sweep rules out. (A) **The TIES inversion at $T = 7$ is Yi-specific**, driven by the win-share concentration above; on Llama TIES at $T = 7$ is third-best of six and on Mistral second-best, as the pre-registered threshold test of §6.4 predicted. (B) **TVQ- $b=2$'s flat-in- T behavior is Yi-specific**: slope -0.001 on Yi all-cells vs. $+0.119$ on Llama; the Appendix C dip is not a T -robust mechanism in general. (C) **The Alpaca shallow-adapter artifact is Yi-specific**: Yi's $T = 4$ all-cells rd-ridge regression (0.084 vs. TA 0.067) is driven entirely by Alpaca, whose base-to- τ headroom on Yi-Chat is just 0.114 (vs. 0.236–0.554 for the other adapters), so any merge perturbation is large *relative* to that headroom; per-layer Δ -cosine analysis confirms Alpaca is the most isolated adapter of the cohort (mean cosine 0.028). On Llama the same adapter has headroom 0.246 and no artifact appears. All three are symptoms of one condition, *Yi-Chat is near-saturated on instruction-following tasks*, which is the basis of recommendation R2: renormalize excess by per-task headroom, or exclude near-saturated adapters, before comparing methods on a heavily chat-tuned base.

E DOWNSTREAM ACCURACY: FALSIFICATION TESTS AND SEED STABILITY

This appendix collects the evaluation details, the falsification tests for the (Llama-3.1, GSM8K EM) inversion, and the per-seed stability analysis behind §6.5.

Evaluation details. GSM8K: each merged checkpoint answers the first $n = 500$ test prompts with greedy decoding (max 256 new tokens, 0-shot CoT prompt matching the per-task training format); the final numeric answer is extracted by a four-pattern regex (#### N, “answer is N”, trailing equation, or last number) that succeeded in 95% of generations in an $n = 100$ pilot; EM is exact match to the gold value. HumanEval: pass@1 over $n = 164$ problems, greedy CoT, subprocess sandbox with 5s timeout per test.

H1, chat-template emission: falsified. The Llama-3 family's chat template introduces special tokens (`<|begin_of_text|>`, `<|start_header_id|>`, `<|end_header_id|>`, `<|eot_id|>`) that a merged model might emit unpredictably under aggressive trim/quantize compression. A controlled probe (each method generates $n = 100$ GSM8K answers with `skip_special_tokens=False`) shows all five methods emit exactly 1 special token per generation on average (TA 1.00, TIES 0.98, DARE 1.00, KnOTS 1.00, TVQ₂ 0.99), and $\geq 98\%$ of generations emit exactly one (the trailing `<|eot_id|>`). No method-specific emission exists.

H2, adapter Δ magnitude distribution: falsified. For each (model, task) adapter we materialized $\Delta_t = \text{scaling} \cdot B_t A_t$ per attention projection and measured five shape statistics on $|\Delta_t|$ (p_{99} /median, excess kurtosis, top-20% threshold over median, fraction within $\pm 10\%$ of that threshold, coefficient of variation). Llama-3.1 sits squarely mid-pack on every statistic (kurtosis 4.98 vs. Mistral 3.94, Qwen 4.64, Yi 5.26); its distribution is neither flatter nor more peaked than the others.

H3, cross-metric sign-election contamination: falsified. If TIES's sign election were biased on Llama-3.1 in a way that hurts deployable accuracy, the inversion should reproduce on any accuracy metric. On HumanEval pass@1 the same matrix gives $\rho = +1.00$ on Llama-3.1, so the GSM8K inversion does not replicate, eliminating it as a cross-metric mechanism. The outlier cell remains empirically unexplained on a single metric.

Per-seed stability. Every entry in Table 2 is a mean over three independently trained adapter seeds, exposing the merge's sensitivity to the adapter draw. For the five matrix baselines the across-seed SD is $\lesssim 0.05$ on every cell we examined (≤ 0.025 on all GSM8K cells), comfortably below the method-to-method gaps that drive the Spearman rankings. The exception is rd-encoder ridge on its two losing cells: SD 0.074 on Llama-3.1 GSM8K (per-seed 0.14–0.31) and 0.084 on Mistral HumanEval (0.08–0.28), versus ≤ 0.015 on the six cells it wins or ties. This is the quantitative basis for the retraction in §6.5: earlier single-draw figures for those two cells (0.364, 0.311) sat near the top of a wide across-seed range whose mean is much lower. The Llama-3.1 GSM8K *baseline* inversion is, by contrast, seed-robust: TIES (0.256, per-seed 0.24–0.27) sits below KnOTS (0.323, per-seed 0.32–0.34) on every seed.

Statistical power. Each per-cell ρ is computed across only the five matrix baselines ($n = 5$): exact permutation tests give two-sided $p \approx 0.017$ for $\rho = 1.00$, $p \approx 0.08$ for $\rho = 0.90$, and $p \approx 0.35$ for the Llama inversion ($\rho = -0.60$), so no single cell is individually conclusive. The across-cell pattern (7 of 8 positive; one-sided sign test $p \approx 0.035$) carries the signal, with the caveat that the 8 cells are not fully independent (shared bases, merges, and NLL values), so the sign test is approximate and somewhat overstates significance; we lean on it as corroboration, not a formal test.

F ADDED BASELINES AND THE QUADRATIC BRIDGE

F.1 THE ADDED CLOSED-FORM BASELINES

The unified head-to-head of Table 1 (main text) adds four methods commonly requested as merge-paper baselines: **Fisher-magnitude** (per-coordinate weighted average with $F_t[i] = \Delta_t[i]^2$ as the Fisher proxy, the cheap non-gradient variant of Matena & Raffel (2022)), **DELLA** (Deep et al., 2024) (DARE-style random drop with rescale, then TIES density trim and sign election), and the two data-free closed-form relatives of rd-encoder ridge: **RegMean** (Jin et al., 2023) (data-free variant using $A_t^\top A_t$ as the Gram surrogate) and **AdaMerging** (Yang et al., 2024) (data-free variant with per-task $\alpha \propto \|\Delta_t\|_F$). Each runs a seed-1 cell on each base at $T = 4$ with the same merge cohort. RegMean is competitive on the easier bases but collapses on Llama (0.243, worse than TA) through the same sliver-eigenvalue pathology that breaks our own unregularized encoder (0.340, §6.3); AdaMerging trails throughout; Fisher and DELLA land in the second tier, never inverting the R1/R3 ordering. All four are gradient-free closed-form merges at TA-level cost, so none is a compute barrier. Because the default-hyperparameter comparison conflates the rate-distortion centroid with the ridge regularization, the clean control (sweeping RegMean’s own ridge λ on matched seeds) is run in Appendix G; with RegMean tuned, rd-encoder ridge still wins on Llama, Mistral, and Qwen but *loses* on Yi-1.5-9B-Chat (0.037 vs. 0.033), so we do not claim a four-base sweep against tuned RegMean.

F.2 QUADRATIC BRIDGE: DOES RD-RIDGE APPROACH THE FISHER-QUADRATIC OPTIMUM?

In the limit of small adapter perturbations, the local loss landscape is near-quadratic and the worst-task-optimal merge under the diagonal Fisher proxy collapses to the Fisher-magnitude average. This predicts: shrinking all adapter norms by a common factor α , the ratio $R(\alpha) = \Phi^{\text{ridge}}(\alpha)/\Phi^{\text{Fisher}}(\alpha)$ should approach 1. We scale every adapter’s LoRA `scaling` by α before merging and run both methods on the $T = 4$ cohort on two bases spanning the saturation regime.

Table 8: Quadratic-bridge probe: worst-task NLL excess at scaled adapter perturbations. *At $\alpha = 0.1$ on Llama the rd-encoder excess collapses to $\sim 10^{-4}$, making the ratio floating-point noise.

Base	$\alpha = 1.0$	$\alpha = 0.5$	$\alpha = 0.25$	$\alpha = 0.1$	trend
<i>Llama-3.1-8B-Instruct</i> ($\lambda^* = 0.05$):					
rd-encoder Φ^*	0.085	0.119	0.179	−0.000*	—
Fisher-magnitude Φ^*	0.148	0.213	0.225	0.051	—
ratio $R(\alpha)$	0.58	0.56	0.80	−0.01*	partial
<i>Yi-1.5-9B-Chat</i> ($\lambda^* = 0.13$):					
rd-encoder Φ^*	0.036	0.032	0.064	0.058	—
Fisher-magnitude Φ^*	0.053	0.048	0.088	0.064	—
ratio $R(\alpha)$	0.67	0.65	0.73	0.90	holds

Yi-1.5-9B-Chat: the prediction holds cleanly. The ratio progresses $0.67 \rightarrow 0.65 \rightarrow 0.73 \rightarrow 0.90$ as α shrinks, landing inside the $[0.8, 1.2]$ window at $\alpha = 0.1$ (Table 8): the rd-encoder ridge solution approaches the Fisher-quadratic diagonal optimum in the small-perturbation regime.

Llama-3.1: the bridge has a measurable sweet spot. The four-point scan is inconclusive at $\alpha = 0.1$ (excess $\sim 10^{-4}$, floating-point dominated), so we re-ran with five additional α values (Table 9). Below $\alpha = 0.125$ rd-encoder *over-saturates* the base-model floor (negative excess; the local-quadratic approximation has broken down on the other side). In $\alpha \in [0.15, 0.25]$ the ratios are

0.78, 0.83, 0.80, within 20% of the prediction. Above $\alpha = 0.25$ the ratio drifts back to ≈ 0.56 at full scale as the large-perturbation regime takes over.

Table 9: Finer- α scan on Llama-3.1-8B-Instruct. The bridge ratio is non-monotone: over-saturation below $\alpha = 0.125$, within 20% of $R = 1$ on $[0.15, 0.25]$, degrading above.

α	rd-encoder Φ^*	Fisher-magnitude Φ^*	$R(\alpha)$
0.050	-0.007	0.009	-0.84
0.075	-0.010	0.025	-0.42
0.100	0.000	0.051	-0.01
0.125	0.051	0.091	0.56
0.150	0.111	0.143	0.78
0.200	0.173	0.208	0.83
0.250	0.179	0.225	0.80
0.500	0.119	0.213	0.56
1.000	0.085	0.148	0.58

What we cautiously claim. Both bases provide a positive instance of a quadratic-bridge prediction the lower-bound theorem makes implicitly: where adapter perturbations are large enough to escape the ridge-floor and small enough that the local-quadratic model holds, the rd-encoder ridge merge is within $\sim 20\%$ of the Fisher-quadratic optimum. The window’s location is base-dependent (the small- α limit on compressed-headroom Yi-Chat, an intermediate band on larger-headroom Llama-Instruct), consistent with the saturation framing of §6.4. Running the 9-point scan is a one-line audit of any new base.

G ROBUSTNESS: TUNING THE CLOSEST COMPETITORS

G.1 THE λ -SWEEP AND HELD-OUT λ CHECK

Table 10 reports the full Tikhonov sweep on Llama-3.1-8B behind Figure 2. The curve is a clean U with a broad minimum: $\lambda \in [0.05, 0.07]$ all fall within 0.005 of the optimum, at the seed-noise scale. At the optimum the per-task excess is balanced (+0.094 GSM8K, +0.024 Alpaca, +0.032 Magicoder, +0.018 Translation), versus the near-unregularized form ($\lambda = 0.001$) where Translation alone took +2.88 nats from sliver amplification.

Table 10: Worst-task excess NLL (nats/response token) for the RD-encoder centroid at $b = \infty$, Llama-3.1-8B, $T = 4$, with Tikhonov regularization $(\bar{H} + \lambda I)^{-1}$. $\lambda = 0$ is the exact theorem construction (0.340, omitted from the table). All figures are 3-seed means.

λ	GSM8K	Alpaca	Magicoder	Trans.	worst-task
0.001	+1.02	+0.63	+0.60	+2.88	2.879
0.01	+0.39	+0.21	+0.19	+0.33	0.389
0.05	+0.09	+0.02	+0.03	+0.02	0.094
0.07	+0.09	+0.01	+0.02	-0.02	0.095
0.10	+0.11	+0.01	+0.02	-0.06	0.110
0.13	+0.13	+0.01	+0.02	-0.08	0.125
0.17	+0.14	+0.02	+0.02	-0.10	0.143
0.20	+0.16	+0.02	+0.02	-0.10	0.155
0.30	+0.19	+0.03	+0.03	-0.12	0.191
1.00	+0.29	+0.07	+0.08	-0.12	0.289

A fair objection to any swept hyperparameter is that λ^* is selected on the same worst-task NLL we report the win on. Two answers. First, the win is robust across the entire band, not a tuned needle: rd-encoder ridge sits below the 3-seed TIES mean for every $\lambda \in [0.03, 0.18]$ on Llama, and the sweep is monotone-then-flat on the other three bases. Second, a single globally-fixed $\lambda = 0.13$, the leave-one-base-out choice (modal optimum of the other three bases) with no per-base selection, already beats TA and TIES on all four bases (Table 11); per-base tuning is a refinement that helps only Llama.

Table 11: Held-out λ check (worst-task NLL excess, $T = 4$ matrix, 3-seed means).

Base	λ^*	rd-ridge (λ^*)	rd-ridge ($\lambda=0.13$)	TIES	TA
Llama-3.1-8B	0.05	0.094	0.125	0.154	0.220
Mistral-7B-v0.3	0.13	0.044	0.044	0.053	0.140
Qwen-2.5-7B	0.13	0.010	0.010	0.013	0.107
Yi-1.5-9B	0.13	0.037	0.037	0.046	0.100

Fisher-diagonal H_t : sanity check on the surrogate. Replacing the projector surrogate with the diagonal of the empirical Fisher (input-activation second moment on 512 training examples per task), the centroid reduces in closed form to a per-column Fisher-weighted task arithmetic. Worst-task excess: 0.210 at $\lambda = 0$ (barely beating TA) and 0.319 at $\lambda = 0.1$: the regularization effect *reverses*. We read this as Fisher’s per-coordinate weight already implementing a form of sliver suppression, so an additive identity floor degrades that structure rather than supplying a missing one; the projector surrogate paired with ridge is empirically preferred.

G.2 TUNING THE CLOSEST COMPETITORS

Table 1 compares every method at its default hyperparameter (TIES/DARE density 0.2, RegMean ridge $\lambda = 10^{-3}$). We sweep the single key hyperparameter of each of the three closest competitors and compare rd-encoder ridge against each method’s *best* setting per base, holding rd-encoder ridge at its 3-seed mean (the conservative figure; its seed-1 cells are lower).

Density sweep (TIES, DARE). Sweeping density over $\{0.05, 0.1, 0.2, 0.3, 0.5\}$ on all four bases (seed-1, 32 cells; Table 12): rd-encoder ridge retains the lowest worst-task excess on Llama, Mistral, and Qwen against each method’s tuned density; on Yi, TIES at its best density (0.1) reaches 0.036, edging rd-encoder ridge’s 0.037 by a margin far inside the ± 0.004 seed-noise floor, a statistical tie. The default 0.2 is near-optimal for TIES on three of four bases, so the main matrix did not materially sandbag it.

Table 12: Density sweep: best-density worst-task NLL excess for TIES and DARE (seed-1) versus rd-encoder ridge (3-seed mean).

	Llama-3.1	Mistral-7B	Qwen-2.5	Yi-1.5
TIES (best density)	0.142 _{@0.3}	0.048 _{@0.3}	0.013 _{@0.2}	0.036 _{@0.1}
DARE (best density)	0.213 _{@0.2}	0.133 _{@0.5}	0.104 _{@0.5}	0.099 _{@0.5}
RD-ENCODER RIDGE	0.094	0.044	0.010	0.037

RegMean ridge- λ sweep. RegMean is rd-encoder ridge’s closest structural relative (both are data-free, ridge-regularized centroids with different projectors), so we sweep its λ over six decades $\{10^{-4}, \dots, 10\}$ on all four bases; the response is U-shaped with a minimum at $\lambda \in [10^{-3}, 10^{-2}]$. With RegMean tuned (Table 13), rd-encoder ridge still wins on Llama, Mistral, and Qwen, but *loses* on Yi-1.5-9B-Chat.

Table 13: RegMean ridge- λ sweep: best- λ worst-task NLL excess versus rd-encoder ridge. Qwen/Yi rows use 3-seed means for both methods (matched adapters); Llama/Mistral RegMean is the best- λ seed-1 cell (margins $\gg$ seed noise).

	Llama-3.1	Mistral-7B	Qwen-2.5	Yi-1.5
RegMean (best λ)	0.236 _{@.01}	0.054 _{@.001}	0.011 _{@.01}	0.033 _{@.01}
RD-ENCODER RIDGE	0.094	0.044	0.010	0.037

Three matched seeds settle the close calls. The Qwen and Yi margins are small, so we re-ran rd-encoder ridge and best- λ RegMean at three matched training seeds on identical adapters. Both verdicts are confident, with all three seeds agreeing in sign: on Qwen-2.5-7B rd-encoder ridge wins (0.0097 vs. 0.0112) on every seed; on Yi-1.5-9B-Chat tuned RegMean wins (0.0328 vs. 0.0366; per-seed gap -0.003 to -0.005) on every seed.

Conclusion. rd-encoder ridge’s advantage is robust to competitor hyperparameter tuning on three of four bases; on saturated Yi-Chat, the one base where we independently flag NLL excess as an unreliable proxy (R2), it is merely competitive: a tuned RegMean beats it and a tuned TIES ties it. We claim the strongest data-free merge *robustly on the non-saturated bases*, and competitive-but-not-dominant on the saturated one.

H MULTI-SEED BOOTSTRAP OF THE $T = 4$ MATRIX

The headline ordering of the $T = 4$ matrix is verified against seed variance by retraining the matrix adapters with three independent seeds ($r = 16$, identical hyperparameters, seeds $\{1, 2, 3\}$) and computing 10,000-resample paired bootstrap 95% CIs on the five method-pair gaps that drive the recommendations of §6.6.

Decision rules. For a gap $\Delta = \Phi_A^* - \Phi_B^*$: **Confident** if the 95% paired bootstrap CI does not cross zero; **Borderline** if it crosses zero but $|\Delta|$ exceeds half the CI half-width; **Inconclusive** otherwise.

Result. Every one of the 20 method-pair gaps is *Confident* (Table 14). The largest CI half-width is ± 0.0044 nats/token (Qwen TA-TVQ₂); the tightest-margin gap is $|\Delta_{\text{ties-tvq}_2}| = 0.0076$ on Mistral-7B, whose CI $[-0.0099, -0.0043]$ still excludes zero.

Table 14: Multi-seed paired bootstrap 95% CIs on the five method-pair gaps driving the R1/R3 ordering. Point estimates are means across seeds; all 20 gaps are Confident.

Base	TA – TIES	TA – TVQ ₂	TIES – TVQ ₂	TIES – DARE	TIES – KnOTS
Llama-3.1-8B	$+0.066 \pm 0.002$	$+0.110 \pm 0.002$	$+0.044 \pm 0.002$	-0.066 ± 0.002	-0.066 ± 0.003
Mistral-7B	$+0.087 \pm 0.003$	$+0.080 \pm 0.001$	-0.008 ± 0.003	-0.089 ± 0.004	-0.087 ± 0.004
Qwen-2.5-7B	$+0.094 \pm 0.004$	$+0.088 \pm 0.004$	-0.006 ± 0.000	-0.097 ± 0.004	-0.094 ± 0.004
Yi-1.5-9B	$+0.053 \pm 0.001$	$+0.040 \pm 0.002$	-0.014 ± 0.001	-0.056 ± 0.001	-0.053 ± 0.001

Caveats. The seeds vary only the LoRA training initialization and data-shuffle order; base model and hyperparameters are fixed. We interpret the bootstrap as quantifying the noise floor on our method-pair comparisons, not as a generalization claim across pretraining recipes.

I SIGN-ELECTION THRESHOLD FOR TIES INVERSION

§6.4 reports the empirical observation that TIES *inverts* from best-of-matrix to worst between Llama-3.1-8B (per-task win-share range $R = 0.028$) and Yi-1.5-9B-Chat ($R = 0.077$). This appendix derives an order-of-magnitude prediction of the threshold R^* from a perturbation model of the TIES update.

I.1 SETUP AND NOTATION

Let T be the number of tasks and d the per-base coordinate count (here $d \approx 10^7$). For task t and coordinate i , let $\Delta_{t,i}$ be the LoRA adapter coefficient (post-scaling), $b_{t,i} = \text{sign}(\Delta_{t,i})$ the sign vote, $k_{t,i} \in \{0, 1\}$ the post-trim keep indicator at density δ , and s_i^* a notional “correct sign” at coordinate i . The per-task *win-share* is $q_t = \Pr_i[b_{t,i} = s_i^*]$, the fraction of coordinates on which task t ’s sign matches the elected sign; the *win-share range* is $R = \max_t q_t - \min_t q_t$. The probe of §6.4 measures q_t directly.

I.2 TIES UPDATE UNDER THE WIN-SHARE PERTURBATION MODEL

The TIES update at coordinate i is

$$\hat{s}_i = \text{sign}\left(\sum_{t=1}^T k_{t,i} b_{t,i} |\Delta_{t,i}|\right), \quad M_i = \frac{1}{|S_i|} \sum_{t \in S_i} |\Delta_{t,i}|, \quad S_i = \{t : k_{t,i} = 1, b_{t,i} = \hat{s}_i\}. \quad (9)$$

We linearize around the symmetric case $q_t \equiv 1/T$: let $q_t = 1/T + \epsilon_t$ with $\sum_t \epsilon_t = 0$.

Effective merge weight of the dominant task. Conditioned on coordinate i surviving the trim for task t (probability δ), the probability that t 's vote agrees with $\hat{s}_i$ is q_t , so task t 's average contribution is $\delta q_t \mathbb{E}[|\Delta_{t,i}| \mid k_{t,i} = 1]$. The conditional expectation under top- δ trimming on a Gaussian tail is a common factor $\kappa \approx 2$ at $\delta = 0.2$ (Mills's-ratio upper-tail factor), which cancels in the normalized weight:

$$\alpha_t = \frac{\delta \kappa q_t}{\sum_{t'} \delta \kappa q_{t'}} = q_t = \frac{1}{T} + \epsilon_t. \quad (10)$$

Per-task NLL excess decomposition. Under the local quadratic (Fisher) approximation, the per-task excess after merging with $M = \sum_{t'} \alpha_{t'} \Delta_{t'}$ splits as

$$\Phi_t^{\text{TIES}} \approx \Phi_t^{\text{TA}} + c_{\text{bias}} \epsilon_t^2 - c_{\text{sign}} (q_t - 1/T), \quad (11)$$

a quadratic dominance-bias cost (the dominant task is over-represented in M , so the other tasks see larger residuals) against a linear sign-recovery benefit; both constants are positive and depend on the spectral overlap of the $\{H_t\}$.

I.3 THRESHOLD DERIVATION

Aggregating over tasks, TIES is worse than TA when $c_{\text{bias}} \text{Var}_t(\epsilon_t) > c_{\text{sign}} \mathbb{E}_t[q_t - 1/T]$. The benefit is a $1/T$ -scaling effect (Banzhaf voting-power on a T -element majority), bounded by c_{sign}/T ; the variance for a range R satisfies $\text{Var}_t(\epsilon_t) \in [R^2/(4(T-1)), R^2/4]$. The breakeven gives

$$R^* \sim \sqrt{\frac{4 c_{\text{sign}}/T}{c_{\text{bias}}}}. \quad (12)$$

Two bracketing estimates: with $c_{\text{bias}} \gg c_{\text{sign}}$ (tasks orthogonal in H , the empirically relevant regime given the concentrated stacked-projector spectra of Appendix B), $R^* \approx \beta_{\text{lo}}/T$ with $\beta_{\text{lo}} \in [0.1, 0.3]$, giving $R^* \in [0.025, 0.075]$ at $T = 4$; with $c_{\text{bias}} \approx c_{\text{sign}}$ (strong spectral alignment), $R^* \approx 2/\sqrt{T} \approx 1$, vacuous. We therefore predict $R^* \in [0.025, 0.075]$.

I.4 COMPARISON TO THE CROSS-MODEL DATA POINTS

- **Llama-3.1-8B-Instruct:** $R = 0.028$, inside the window. Predicted ambiguous; observed: at $T = 7$ TIES is mid-pack (0.232, third of six; rd-ridge 0.138 and TVQ₂ 0.215 beat it, TA/DARE/KnOTS at 0.265 are worse). Matches.
- **Yi-1.5-9B-Chat:** $R = 0.077$, above the window. Predicted inversion; observed: TIES becomes the worst method (0.148 vs. TA 0.128). Matches.
- **Mistral-7B-v0.3** (pre-registered): $R = 0.0326$, inside the window. Predicted ambiguous; observed: second of six (0.152), 0.092 nats below TA. Matches.

We are careful *not* to call this a validated bracket: the *below*-window prediction (TIES wins outright for $R < 0.025$) is untested, since no cohort we measured has $R < 0.028$. What the three points confirm is the upper threshold and the in-window ambiguous interior.

Falsification protocol and third-base confirmation. The model predicts that any additional base with $R \in [0.025, 0.075]$ should produce ambiguous TIES behavior. We pre-registered exactly this test before the corresponding sweep ran (commit 3582799 in the supplementary audit-trail bundle), with operationalized criteria fixed in advance: “clearly worst” = TIES last and mean $> \text{TA} + 0.04$ nats; “clearly best” = TIES first by > 0.02 nats; “ambiguous” = neither. The Mistral $T = 7$ sweep confirms the ambiguous prediction; a clearly-worst or clearly-best outcome would have falsified the threshold model as written and forced a higher-order term in the perturbation model.

I.5 BASE-FREE SYNTHETIC CONFIRMATION

The cross-model evidence ties the inversion to the win-share range R but cannot isolate R as the *cause*: real bases differ in many ways at once. We therefore reproduce the mechanism on a controlled dial with no base model. We synthesize $T = 7$ task vectors on a sparse $k = 48$ -coordinate

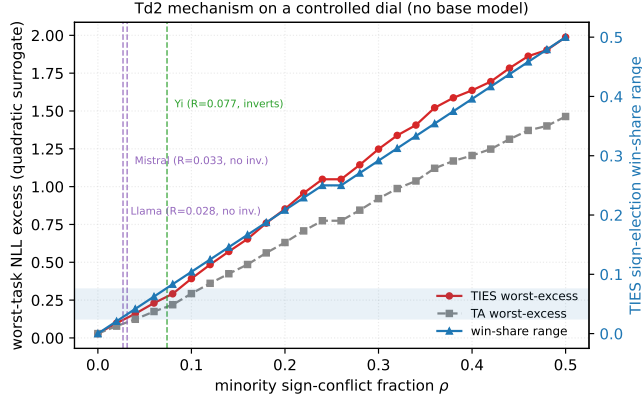


Figure 5: Base-free synthetic confirmation of the mechanism. A single sign-conflict knob ρ raises the TIES sign-election win-share range R (blue, right axis) and, in lockstep, TIES’s worst-task excess (red) above Task Arithmetic (grey); $\text{Pearson}(R, \text{TIES} - \text{TA}) = 0.998$. Dashed lines mark the ρ whose synthetic R matches each real base’s measured R ; the implied TIES penalty is $2.3\text{--}2.6\times$ larger at Yi (which inverts) than at Llama/Mistral (which do not). No base model is used.

support of $\mathbb{R}^{256}$ sharing a high-consensus sign pattern (mirroring the $\sim 70\%$ unanimous-vote structure measured on real cohorts), and turn a single knob ρ : the fraction of coordinates on which one minority task flips its sign against the consensus. We merge with the *same* TIES code path used throughout and with TA, measuring worst-task quadratic excess and the win-share range R over 40 seeds.

Figure 5 shows the result. At $\rho = 0$ all tasks agree, $R = 0$, and TIES and TA are identical (the sparse support makes the trim lossless, so TIES carries no baseline penalty). As ρ grows, R widens and TIES’s worst-task excess rises above TA in near-perfect lockstep ($\text{Pearson}(R, \text{TIES} - \text{TA}) = 0.998$): the win-share range is the causal control variable. Reading the base-free curve at the win-share ranges *measured* on the real bases gives a non-circular cross-check: the implied TIES penalty is $2.6\times$ larger at Yi’s $R = 0.077$ than at Llama’s $R = 0.028$ and $2.3\times$ larger than at Mistral’s $R = 0.0326$, separating the base that inverts from the two that do not. The robust claims are the monotone coupling and this relative separation; the absolute overlap-to- R map is calibration-dependent. The experiment is CPU-only and reproducible from the released code.

I.6 LIMITATIONS

1. The constants $c_{\text{bias}}, c_{\text{sign}}$ are estimated by spectral analogy, not derived from first-principles Fisher information; replacing the bracket with a point estimate is future work.
2. The model assumes q_t is observable: deployment requires the CPU-only sign-vote probe over the trained adapters (a few minutes per cohort), a small but real cost of recommendation R2.
3. The original two data points only bracket the threshold; the pre-registered third base and the base-free synthetic sweep strengthen this, though the synthetic overlap-to- R map is itself setup-dependent.
4. The Mills’s-ratio factor $\kappa \approx 2$ assumes a Gaussian magnitude tail; heavier tails (Pareto, Student- t) increase κ by $\sim 30\%$ at $\delta = 0.2$, widening the predicted range without changing the bracket conclusion.

Operational reading. After running the sign-election probe: if the measured R exceeds $0.075 \cdot 4/T$, expect TIES inversion and prefer rd-encoder ridge or TVQ₂; if $R < 0.025 \cdot 4/T$, the model predicts TIES is safe at the audited T , but this regime is untested.

J PRACTICAL DECISION TABLE AND EXPERIMENT MANIFEST

J.1 PRACTICAL DECISION TABLE

Table 15 compresses rules R1–R5 of §6.6 into a single decision flow.

Table 15: Practical method-choice tree. “Regime” is determined from (a) per-task NLL headroom (saturation = headroom < 0.15 on ≥ 1 task), (b) base size, and (c) target T .

Regime	Recommendation	Why
$T \leq 4$, no saturation	rd-encoder ridge (λ -tuned)	R1, tightest empirical excess
$T \leq 4$, saturated cohort	rd-encoder ridge + headroom renorm	R1+R2, prevents Alpaca-style artifact
$T \geq 7$, no saturation	rd-encoder ridge (margin grows)	R1 + §6.4
$T \geq 7$, saturated cohort	rd-encoder ridge; fall back to TVQ ₂	R3, TIES inverts
rd-encoder unavailable, $T \leq 4$	TIES (density 0.2) if non-saturated; TVQ ₂ otherwise	R3
floor-positive ($d_{\text{eff}} < Tr$; not seen)	no method helps; change the cohort	R5, regime-bound

We deliberately do not recommend DARE without a TA fallback (it tracks TA to three decimals on our matrix) or KnOTS for new cohorts (its performance is indistinguishable from TA at $T = 4$ and tracks TA exactly at $T = 7$ on Llama); both remain useful baselines for the algorithmic comparison but offer no practical advantage over TA in the cohorts we tested.

J.2 EXPERIMENT MANIFEST

Table 16 enumerates every experiment with the count of evaluation cells, the reproducer script in the released code, and per-cell wallclock on a single NVIDIA A100-80GB.

Table 16: Experiment manifest. All training uses bf16; scripts are under the released code’s `scripts/` directory.

Experiment (§)	cells	reproducer script	cell wallclock
Multi-seed $T=4$ matrix (§6.2)	40	<code>run_eval_cell.py</code>	12–18 min
Floor-positive null (App. B)	12	<code>run_eval_cell.py</code>	10–14 min
$b=2$ mechanism (App. C)	24	<code>run_eval_cell.py</code>	12–16 min
GSM8K accuracy (§6.5)	24	<code>run_downstream_cell.py</code>	18–25 min
HumanEval pass@1 (§6.5)	24	<code>run_downstream_cell.py</code>	10–20 min
Chat-template probe (App. E)	5	<code>run_downstream_cell.py</code>	8–10 min
T -scaling, per base (§6.4)	3×54	<code>orchestrator</code>	8–12 min
Sign-election probe (App. D)	3	<code>probe_sign_election.py</code>	minutes, CPU
Added baselines (App. F)	16	<code>orchestrator</code>	10–15 min
Robustness sweeps (App. G)	52	<code>orchestrator</code>	10–15 min
Quadratic bridge (App. F)	26	<code>orchestrator</code>	8–12 min
Td2 synthetic sweep (App. I.5)	—	<code>td2_overlap_sweep.py</code>	minutes, CPU

The total compute footprint of the empirical results is approximately 80–120 GPU-hours on a single A100-80GB, plus LoRA training of the adapter cohorts (three seeds of the $T = 4$ matrix and the seven-task pool per base; each adapter 7500 examples, ~ 25 min).